

Network Intrusion Detection Algorithm and Simulation of Complex System in Internet Environment

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Abstract—This paper proposes a CNN-BiLSTM intrusion detection model for complex system networks. The model performs data over-sampling on the unbalanced data set, which reduces the gap in the amount of category data. It is based on the integration, cooperation, and selectivity of methods and mechanisms in the intrusion detection system, so as to achieve the idea of optimization. In the intrusion detection system, an intrusion detection system based on a variety of detection methods and technologies is proposed, and an integrated, cooperative, and selective overall structure is established. It will be based on distributed intrusion detection and feature engine analysis of intrusion detection, efficiency an increase of 6.7%.

Keywords—Network Intrusion, Intrusion Detection, Complex System, Internet Environment

I. INTRODUCTION

With the rapid development of science and technology, mankind has entered the information age. In this era, computer technology and network technology have penetrated into all fields of society. In particular, the appearance has brought new feelings to people's daily life. However, these technologies, like other new technologies, have advantages and disadvantages. While we are enjoying the convenience and speedy convincing they bring, security issues also quietly come to us. The frequent occurrence of various network information security incidents has sounded a security alarm for people, such as DDoS attacks and XSS attacks. These security issues have led to the emergence of detection technologies, such as the construction of firewalls, malicious code detection, and intrusion detection [1-6].

As the most basic protection of the computer, the firewall is set between the protected network and the external network to protect the internal network information from illegal intrusion from the external network. It is a passive defense and is affected by various restrictions and areas. Therefore, when dealing with external attacks There are certain restrictions. Malicious code detection environment is safer and faster, but it is susceptible to the influence of wrapper and obfuscation technology. The shell needs to be decrypted before analysis. Intrusion detection is a supplement to the firewall. It can actively defend against network security threats. By analyzing the data in the network and the system to determine whether there is intrusion, it can protect the system and make up for the shortcomings of traditional defense methods. It is an important means to discover abnormal network behaviors, so it is of

great significance to strengthen the research of intrusion detection methods. With the development and application of information technology, the connotation of information security is constantly extending, from the initial confidentiality of information to the integrity, availability, controllability and non-repudiation of information, and then to "attack (attack), prevention (prevention), measurement (detection), control (control), management (management), evaluation (evaluation)" and many other theories and technologies [7-14].

In the field of information security research, it is necessary to comprehensively utilize the long-term knowledge accumulation and the latest development results of many disciplines such as mathematics, physics, communication and computer, conduct independent innovation research, strengthen top-level design, and propose systematic, complete, and collaborative solutions. Compared with other disciplines, information security research emphasizes independence and innovation, and innovation can resist various attacks and adapt to the needs of technological development. At present, people's focus in the field of information security mainly includes the following aspects: (1) crypto theory and technology; (2) security protocol theory and technology; (3) security architecture theory and technology; (4) information confrontation theory and technology Technology; (5) Network security and security products. Anomaly detection is also called behavior-based detection, which is realized by observing the difference between the current activity and the historical normal activity of the system. Anomaly detection first establishes a state model of the system's normal activities and continuously updates it, and then compares the user's current activities with this normal model. If the difference between the two exceeds the set threshold, it is considered that an illegal attack has occurred. Misuse detection analyzes various known types of attack methods, and finds out the set of possible "attack characteristics". Misuse intrusion detection uses these feature sets or corresponding rule sets to preprocess the current data source, and then perform feature matching or rule work. If a match that meets the conditions is found, it is considered that an attack has occurred. The key issue of misuse detection is how to correctly represent the attack signature [15-21].

In particular, the attack signature must be able to accurately represent the intrusion behavior and all possible

variants, while not including non-intrusion behavior. In fact, what an administrator really cares about is whether the intrusion detection system is suitable for its own operating environment, and hopes to find the best combining point between the various performances of the intrusion detection system, so that it can better meet the actual needs. In response to this situation, this paper designs and implements a network intrusion detection system that can interact with system administrators [22-24].

II. THE PROPOSED METHODOLOGY

A. Complex System Network

The optimization control of complex systems is one of the hot spots of research work in recent years. The traditional hierarchical control of large-scale dynamic systems is obtained under the premise of accurate models. However, the mathematical models of many actual systems are only some kind of actual approximation. It is difficult to obtain an accurate model, so it is difficult to obtain the optimal control of the actual system. In the case of differences between the actual and the model, Roberts first proposed the integrated dynamic system optimization and parameter estimation (DISOPE) method. This method solves the model-based optimal control problem and parameter estimation problem through iterative solutions to obtain the true optimum of the actual problem untie.

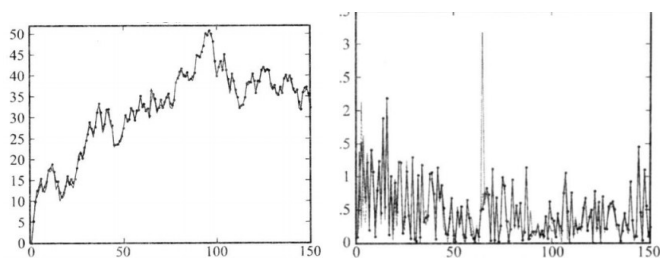


Fig. 1. Complex System Network

The BP neural network is determined by the number of network layers, the number of nodes, activation functions, initial weight coefficients, learning algorithms, and system errors. To determine these requires certain principles: Based on previous experience, priority is given to the 3-layer BP neural network: output Layer, input layer, hidden layer. Under the premise of ensuring accuracy, the goal is to minimize the number of hidden layer nodes. The hidden layer is related to many factors, such as the characteristics of the sample data and the type of the transfer function, the number of input and output nodes. The hierarchical control method of the nonlinear dynamic large-scale system is combined with the integrated method of dynamic system optimization and parameter estimation. Form the DISOPE algorithm of nonlinear dynamic large-scale system.

Becerra and Roberts proposed a hierarchical DISOPE algorithm for continuous nonlinear dynamic large-scale systems, and the article proposed two hierarchical optimization methods for nonlinear discrete dynamic large-scale systems to solve the optimal nonlinear discrete dynamic large-scale systems. control. First, the data is modularized, and then these data modules are distributed to various machines for parallel processing. The processing process between them

is not related, so the processing efficiency will be greatly improved. Starting from the model, through iterative calculations, the actual system can be obtained with the model-the real optimal solution when the actual difference exists, and the correlation between the subsystems can be a non-linear correlation.

B. Complex System Network Intrusion

Anomaly detection is also called behavior-based detection, which realizes detection by observing the difference between the current activity and the historical normal activity of the system. Anomaly detection first establishes a state model of the normal activity of the system and continuously updates it, and then compares the user's current activity with this normal model. If the difference between the two exceeds the set closed value, it is considered that an illegal attack has occurred.

$$I_N(f) = \sum_{i=1}^{2n} w_i f(\xi_i) \quad (1)$$

$$f(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

Intrusion detection (Intrusion Detection) refers to the collection of information from the network or host, such as security logs and audit records, and detection and analysis of whether there is any violation of security policies. Intrusion detection system (Intrusion Detection System, IDS) is a security management system for computer systems and networks. As an active and effective active security defense technology, it can monitor the real-time traffic in the network and determine whether any abnormal behavior enters the system. If so, an alarm will be issued to prevent the entry of abnormal behavior, thereby ensuring the confidentiality, integrity and availability of the system or network resources. The host-based intrusion detection system collects and analyzes log files, port information, and device types of the host, and provides visual analysis and policy management of different types of intrusions.

The intrusion detection system has high detection efficiency and fast speed. It detects the use of the host at all times, and basically sends an alarm before the host is attacked. However, the system cannot obtain and detect network data. The host-based intrusion detection system (HIDS) extracts data (such as audit records, etc.) from a single host as the data source for intrusion analysis, while the network-based intrusion detection system extracts data from the network (such as network link layer data frames) As a data source for intrusion analysis. Generally speaking, a host-based intrusion detection system can only detect a single host system, while a network-based intrusion detection system can detect multiple host systems in this network segment, and multiple network-based intrusion detection systems distributed on different network segments Can work together to provide stronger intrusion detection capabilities. Neural network has a high degree of learning and self-adaptive ability. Through continuous training and learning of input normal samples and

abnormal samples, the neural network can not only identify the known intrusion behavior features in the training samples with high accuracy, but also identify new intrusion behavior features and existing intrusion behaviors with a certain probability. Know the variants of intrusion

C. Complex System Network Intrusion Detection Algorithm and Simulation

The attack behavior in the network is not completed at a single point in time, but will be sent continuously for a period of time, and the intrusion behavior can be better detected by analyzing the correlation performance of the time series before and after the network data. With reference to the advantages of CNN and LSTM in intrusion detection, this paper proposes an intrusion detection model that combines convolutional neural networks with bidirectional long and short-term memory networks and optimizes the loss function to improve the accuracy of intrusion detection and reduce the false alarm rate.

$$u(t) = \int_0^t \frac{(t-s)^{a-1}}{\Gamma(a)} ds - c_1 \quad (3)$$

$$\frac{Q}{Q_{\max}} = f\left(\frac{1}{L}\right) \quad (4)$$

The model first takes advantage of CNN's ability to extract deep local features to effectively extract sample data, and then uses a bidirectional long-term short-term memory network to learn and extract correlation features between continuous data in both positive and negative directions. Finally, it is classified. The model comprehensively considers the time and space correlation of the intrusion data, can mine the unknown features and internal dependencies between the data, reduce the false alarm rate, and improve the detection effect. However, the existing CNN-BiLSIM algorithm models are mostly used in speech recognition, text analysis and other fields, and they have not yet been widely used in the field of network intrusion detection. The main idea of the proposed MRGA-BP average method algorithm is: In the Map stage of the GA algorithm, N individuals are randomly generated, uploaded to the HDFS file system, and the value of each individual is read. Each individual represents the value of each BP network. Weight, call the BP neural network algorithm, the input weight of each individual and the value of each sample, perform the BP neural algorithm, and find the error sum of each individual corresponding to all samples. This error sum is called the global error value. The error value is used as the output value of the Map stage of the GA genetic algorithm, and the input value of the Redcuc stage is the output value of the Map. The fitness of each individual is calculated, followed by the selection, crossover, and mutation of the genetic algorithm.

After several iterations, the optimal individual is selected and output to the HDFS file as the initial weight of the BP neural algorithm. The intrusion detection system can also monitor the information transmitted in the network to discover the existence of network-based attacks.

III. EXPERIMENT

The intrusion of the complex system network model is shown in the figure.

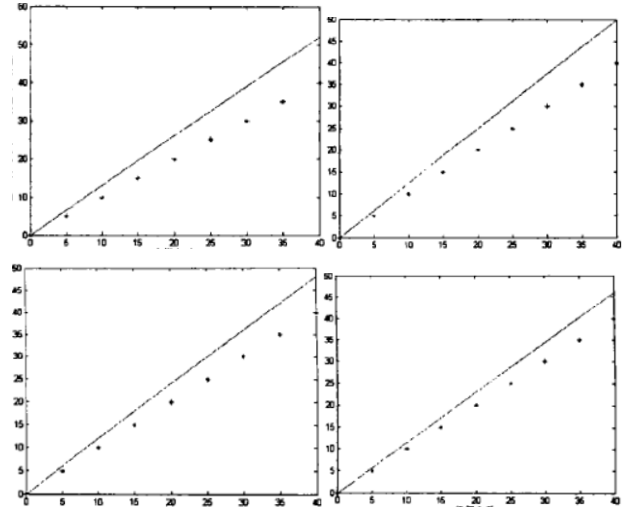


Fig. 2. Complex system network model

The application of complex network intrusion detection algorithm is shown in the figure.

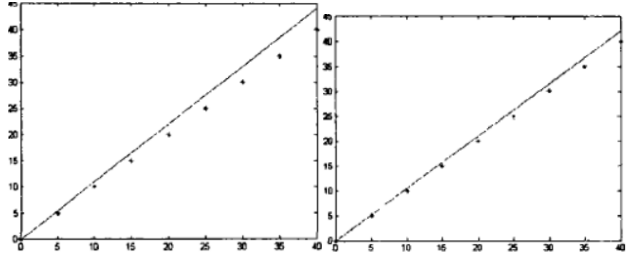


Fig. 3. Application of Intrusion Detection Algorithm in Complex Network

The intrusion detection mechanism is shown in the figure.

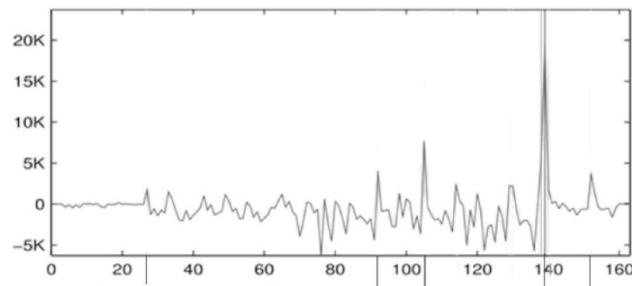


Fig. 4. Intrusion detection mechanism

IV. CONCLUSION

The intrusion detection system is an important security auxiliary system and an important part of the PPDR model. As a proactive security protection technology, intrusion detection provides real-time protection against internal attacks, external attacks and misoperations, intercepting and responding to intrusions before the network system is compromised. Intrusion detection plays a very important role in the network security protection system. It solves the requirement that the

traditional network security model cannot adapt to the increasingly developed network technology. It is an important part of the PPDR model and a practical and usable network security system.

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